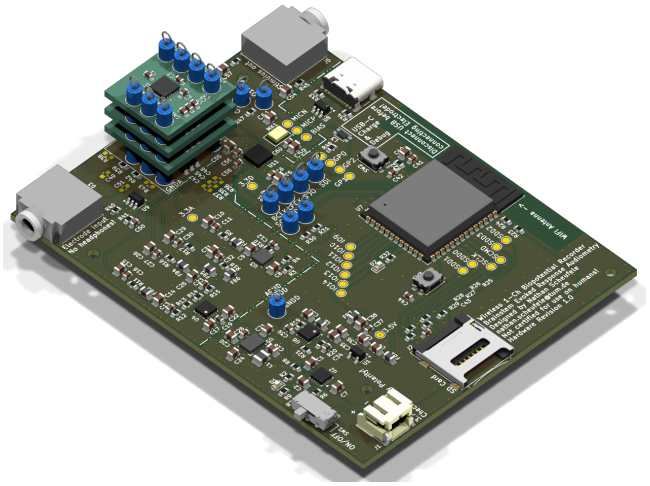
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Development of a Portable, Ultra-Low-Noise Biosignal Acquisition System for Brainstem Evoked Response Audiometry

Forschungspraxis (Master) – Technical Report

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Abstract *Wireless BERA* is a battery-powered, single-channel biopotential recorder for auditory brainstem response (ABR) audiometry that is operated from a web browser over Wi-Fi. One audio codec (TAC5212) digitises the amplified electrode signal and at the same time generates the acoustic stimulus, so stimulus and acquisition share one sample clock. In this Forschungspraxis the board was designed, manufactured as a 4-layer PCB, hand-assembled, brought up and given its firmware. It was then validated in vivo: a chest ECG agrees beat by beat with an Apple Watch (RR difference $0.2 \pm 1.8\,\mathrm{ms}$), the driven right leg lowers mains pickup by 10 to $18\,\mathrm{dB}$, and a click-evoked brainstem response with a wave-V candidate at $6.1\,\mathrm{ms}$ was recorded from the author ($p = 0.001$, permutation test), absent in an earbud-out control.

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1 Motivation and Goals

Brainstem evoked response audiometry records the potentials that the auditory nerve and brainstem produce within about $10\,\mathrm{ms}$ after an acoustic click. Wave V, the most robust peak, is a few hundred nanovolts large and sits on microvolts of EEG, muscle activity and mains interference. It only becomes visible after averaging thousands of stimulus-locked sweeps [1, 2]. Clinical systems are bulky, mains-connected and expensive.

The aim of this work was a compact, battery-powered alternative with

- a low-noise differential front end with programmable gain,
- active common-mode suppression by a driven right leg (DRL) [3],
- an acoustic stimulus output on the same clock as the acquisition,
- battery operation, isolated from mains while recording,
- wireless data transfer and operation from any browser.

The emphasis was on the hardware; the firmware was written to bring the board up and to make first measurements possible.

2 System Overview

The signal paths of the board (Figure 1) are:

- **Biopotential:** electrode connector J6 → LTC6373 instrumentation amplifier → TAC5212 ADC channel 2,
- **Sound:** microphone → ADC channel 1,
- **Stimulus:** TAC5212 DAC → connector J11 (insert earphone),
- **Common mode:** OPA376 DRL driver → third electrode,
- **Data:** $\mathrm{I^2S}$ → ESP32-S3 → Wi-Fi → browser.

Because ADC and DAC run on the same bit clock, every click falls on a fixed sample of the recording. This is the prerequisite for stimulus-synchronous averaging and was used directly in the ABR measurements (Section 6.2).

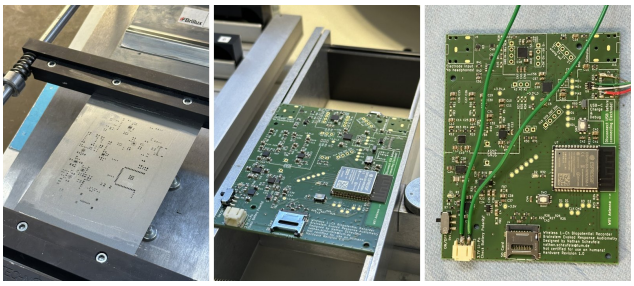


Figure 1: Assembly of v1.0 by hand: stencil printing of solder paste, placed components before reflow, and the assembled board with the battery leads.

3 Hardware Design

3.1 Analog Front End

Instrumentation amplifier. The LTC6373 [4] is a fully differential instrumentation amplifier with pin-selectable gains from 0.25 to 16. It runs from clean $\pm 5\,\mathrm{V}$ rails; three ESP32 GPIOs select the gain. Its output common mode is set to

$$V_{\mathrm{OCM}} = 3.3\,\mathrm{V} \cdot \frac{10\,\mathrm{k\Omega}}{14.99\,\mathrm{k\Omega}} \approx 2.20\,\mathrm{V}, \tag{1}$$

so that the output can drive the codec directly. The electrodes are AC-coupled (C49/C52) and biased through $1\,\mathrm{M\Omega}$ to the DRL reference.

ADC and DAC. The TAC5212 [5] samples both channels with 24-bit resolution at 8 to $96\,\mathrm{kHz}$ and has a two-channel DAC. Full scale is $2\,\mathrm{V}$ rms; at gain 16 one 16-bit step of the data stream corresponds to $5.4\,\mu\mathrm{V}$ at the electrodes.

Driven right leg. An OPA376 senses the input common mode and drives it back, inverted, into the third electrode through $110\,\mathrm{k\Omega}$, which limits the current to below $50\,\mu\mathrm{A}$.

Stimulus output. Both DAC outputs are used single-ended, AC-coupled through $22\,\mu\mathrm{F}$ and $10\,\Omega$ into a $3.5\,\mathrm{mm}$ jack for insert earphones.

3.2 Power Supply

A single Li-ion cell is charged from USB-C by a BQ24074 with power-path management (Figure 2). A TPS63000 buck-boost supplies $3.5\,\mathrm{V}$ for the ESP32-S3 and the microSD card. A TPS65132 charge pump generates $\pm 5.4\,\mathrm{V}$, from which the TPS7A39 dual LDO [6] derives the $\pm 5\,\mathrm{V}$ analog rails. The codec has

its own ultra-low-noise 3.3 V analog (TPS7A94) and digital (TPS7A2033) regulators. All analog rails are fed in a star through ferrite-bead LC filters. Recordings are meant to be made on battery only, which the silkscreen states next to the USB connector.

3.3 Digital Section and PCB

An ESP32-S3-WROOM-1-N16R8 [7] (16 MB flash, 8 MB PSRAM) provides Wi-Fi, USB and a microSD slot. It is the I^2S clock master; the codec locks its PLL to the bit clock. The board is a 4-layer KiCad design manufactured by Aisler, with separate analog and digital grounds joined at one point and the front end placed next to the electrode connector, away from the radio and the switching regulators.

A plug-in daughterboard carries a further LTC6373 at gain 16; several can be stacked (title page) so that their uncorrelated noise averages down by about $\sqrt{n}$.

The analog design was simulated in LTspice before layout: frequency response, input-referred noise, the transient response to a modelled ABR train, and the stability of the DRL loop.

4 Manufacture and Bring-up

The PCB was assembled by hand with a stencil and reflow (Figure 1). All rails came up as designed (codec AVDD 3.31 V, charge pump ±5.4 V), the codec and the charge pump answer on I^2C, and the codec PLL locks at all sample rates from 8 to 96 kHz.

Two deviations from the schematic remain on the prototype. The gain-select resistors R35–R37 are not fitted, so the amplifier runs at its default gain of 16, which is the gain intended for ABR anyway. The MEMS microphone is not fitted either; instead the electret microphone of a 4-pole earbud headset was wired to the microphone input, and its left earbud to DAC channel 2 as the stimulus transducer.

5 Firmware

The ESP-IDF firmware brings the board up in a fixed order (I^2S clock before codec configuration) and serves a single web page as the complete user interface, so no software has to be installed. It provides

- acquisition at 8 to 96 kHz, decimated and streamed over a WebSocket, with a sequence number and the absolute frame index in every packet,
- on-device RMS, peak, DC and clipping meters,
- a stimulus generator (click, tone burst, chirp, sine, noise) with selectable rate, width and alternating polarity,
- battery, rail and codec monitoring and a codec register interface,
- Wi-Fi as access point and client at the same time, plus a serial console.

For the measurements, small Python tools record the stream to disk (with lost packets kept as gaps on the frame grid) and do the averaging and statistics offline. A live ECG view, an audio recorder and a short audio check for the earphone were added to the web interface afterwards; they build but have not yet been tested on the board.

6 Measurements

All recordings were made on the author on 16 October 2026 with the board on battery and the laptop disconnected from mains. Disposable foam electrodes (Philips 40493) were used. Values are referred to the electrodes.

6.1 ECG and Driven Right Leg

A chest lead (left lower rib cage against right clavicle) served as a large, well-known reference signal and as a test of the DRL. Four 60 s recordings at 1 kHz compared the DRL loop open, closed through an electrode on the right lower rib cage, and closed by holding the DRL lead in the hand (Table 1, Figure 3).

Table 1: Mains pickup in the chest ECG.

DRL	50 Hz µV rms	150 Hz µV rms	**change** at 50 Hz
open	1556	573	–
lead held in hand	191	43	−18 dB
chest electrode	498	292	−10 dB
chest, repeated	383	226	−12 dB

The DRL works: closing the loop lowers mains pickup by 10 to 18 dB. The hand contact suppresses most, but adds forearm EMG. The QRS complex (about 1.7 mV) is resolved in every recording without clipping. The averaged beat is a normal sinus beat: PR 155 ms, QRS 90 ms, QTc 375 to 391 ms. The codec's built-in 1 Hz high-pass filter, active by default, flattens the T wave slightly; it can be switched off over the register interface.

In a further recording the board ran in parallel with an Apple Watch ECG (lead I, 512 Hz). After aligning the two by their RR sequences, all 30 RR intervals of the watch recording agree within 0.2 ± 1.8 ms (Figure 3c), including the respiratory sinus arrhythmia between 800 and 1060 ms.

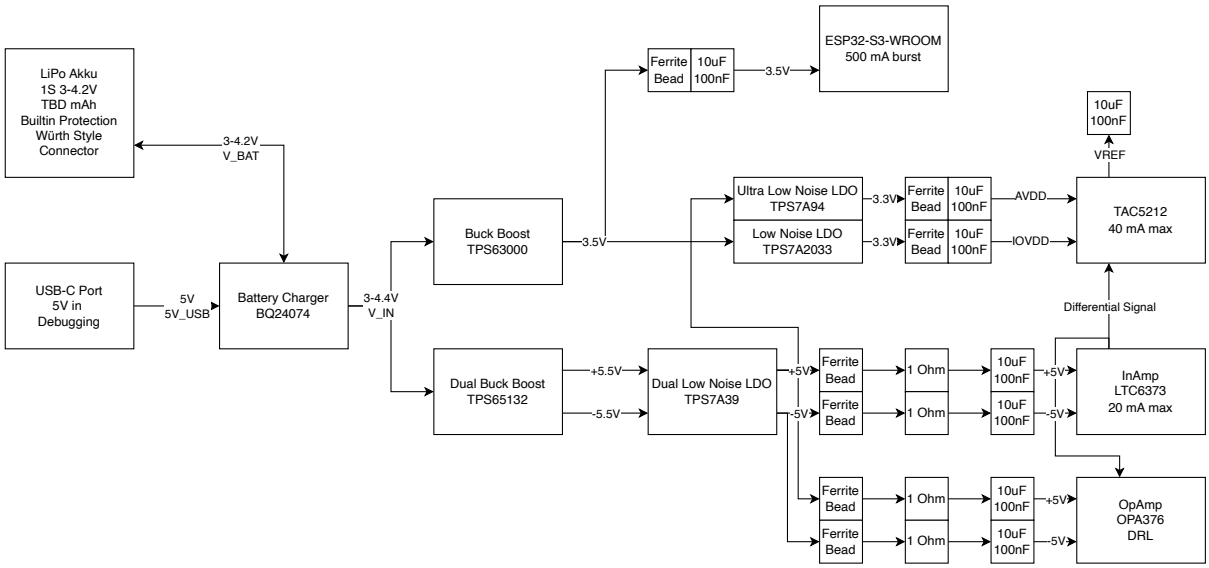


Figure 2: Power architecture of Wireless BERA v1.0.

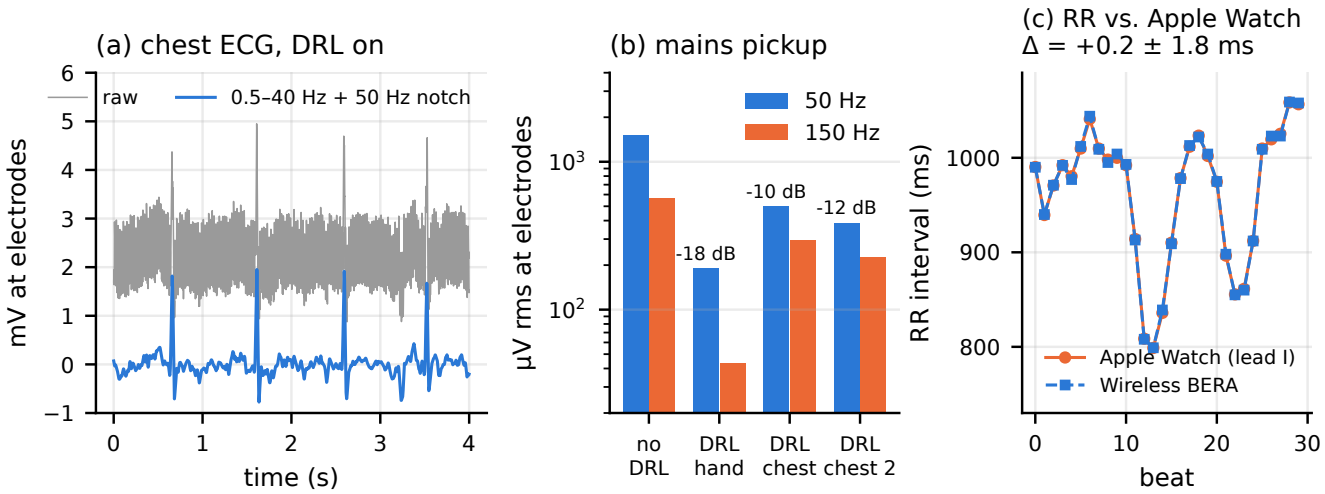


Figure 3: (a) Chest ECG with DRL, raw (offset) and filtered. (b) Mains pickup for the four DRL configurations. (c) Beat-by-beat RR intervals of Wireless BERA and an Apple Watch recorded at the same time.

6.2 Auditory Brainstem Response

Setup. Electrode+ was placed at the high forehead, electrode− on the left mastoid and the DRL on the right mastoid (Figure 4). Rectangular 105 µs clicks were presented to the left ear at 21.1 s^{-1} with alternating polarity. The rate is deliberately not a submultiple of 50 Hz: the mains phase then rotates from sweep to sweep (by 133°) and averages out, whereas at 20 s^{-1} or 25 s^{-1} the harmonics would add up coherently. The stream ran at 24 kHz (48 kHz with 2:1 decimation), where about 1.1 % of packets were lost; at 48 kHz the Wi-Fi link lost about 11 %.



Figure 4: Electrode− on the left mastoid. The earbud cable (black) passes right over the electrode; it couples the electrical click artefact into the recording.

Processing. The continuous recording was band-pass filtered (100 to 1500 Hz, zero phase) and notch filtered at all 50 Hz harmonics, then cut into sweeps from −2 to 12 ms. Time zero is the electrical click artefact, found from the alternating-polarity difference. Sweeps above ±25 µV were rejected in ± pairs, so that artefact and cochlear microphonic keep cancelling, together with restless 64-sweep blocks. Significance was tested by permuting the signs of whole ± pairs (1000 permutations), which keeps the artefact cancelled but destroys any stimulus-locked response.

Results. Four runs with six stimulus conditions were recorded (Table 2). Run 2, lying down with click amplitude 0.05 of DAC full scale, shows a significant stimulus-locked response with a positive peak at 6.1 ms (bootstrap 95 % CI 5.8 to 6.3 ms) and about 160 nV amplitude (Figure 5). Both click polarities show it, so it is neither stimulus artefact nor cochlear microphonic, and its latency fits wave V at a moderate level. In the control run the earbud was taken out of the ear and placed next to the head, with clicks of similar level (0.035): the electrical artefact remains, the peak at 6 ms does not ($p = 0.46$). Run 1 shows the peak at the same latency but stays below significance, as do the runs with fainter clicks and a tone burst.

Table 2: ABR runs (21.1 s^{-1}, left ear). Noise: sweep SD range over 64-sweep blocks.

Run	sweeps	noise µV	p
1 click 0.10, seated	3904	2.9–4.1	0.24
2 click 0.05, lying	5504	1.9–2.2	**0.001**
3 click 0.035	4416	4.7–7.0	0.89
3 sham (no sound)	3840	4.1–5.9	0.76
3 tone burst 2 kHz	4224	3.2–4.7	0.12
4 earbud out (control)	4608	3.5–5.5	0.46

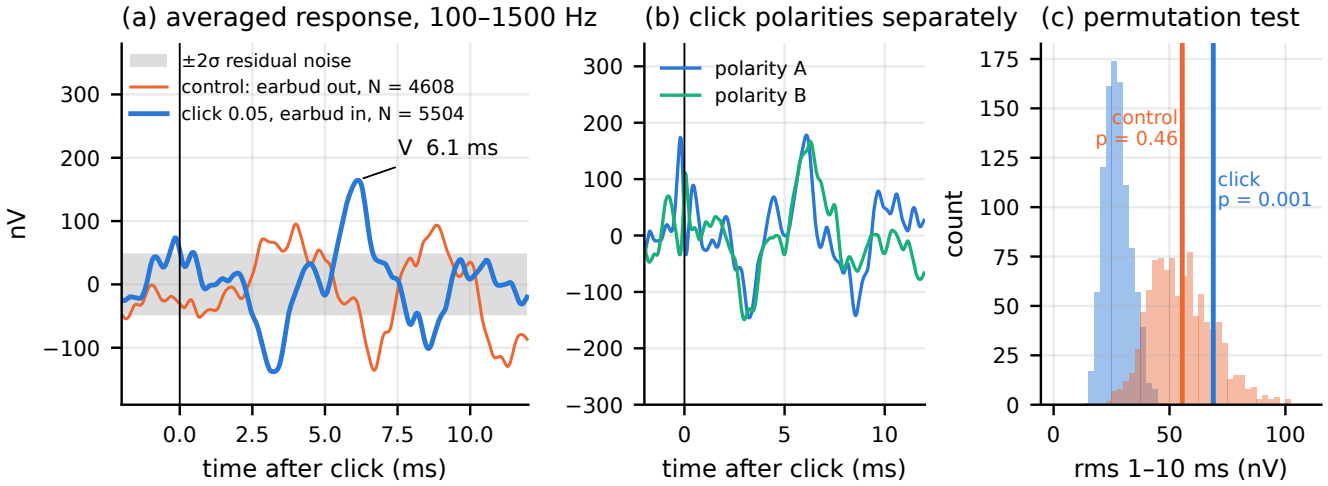


Figure 5: (a) Averaged response of run 2 (blue) with wave-V candidate, and of the earbud-out control (orange); grey: ±2σ of the residual noise of run 2. (b) Run 2 averaged separately for the two click polarities. (c) Permutation null distributions and observed values.

6.3 What Limits the Noise

The noise per sweep, not the electronics, decides whether a run shows a response. Over all runs it follows the EMG power of neck, jaw and forehead muscles closely ($R^2 = 0.86$ on a log scale, Figure 6). In run 2 it fell from 20 µV to 2.2 µV once the subject had settled, after about 130 s. Above 1.5 kHz, where little biological signal remains, all runs share the same floor of about 32 nV/√Hz, close to the expected amplifier and quantisation noise.

The other interference sources turned out to be negligible:

- **Mains:** 0.2 to 6 µV at the head with DRL, against 1.5 mV at the chest without; no harmonic is detectable after the notch filters.
- **Wi-Fi:** no artefact at the packet rate or the beacon interval (below 30 nV). The packet losses fall on a fixed 524 ms grid of the Wi-Fi link and only cost about 4.5 % of the sweeps.
- **Heart:** no ECG is detectable at the head (below 10 to 15 µV); its residual in the average is below 1 nV.
- **Battery:** no change in noise or spurs as the cell fell from 3.42 to 3.23 V.

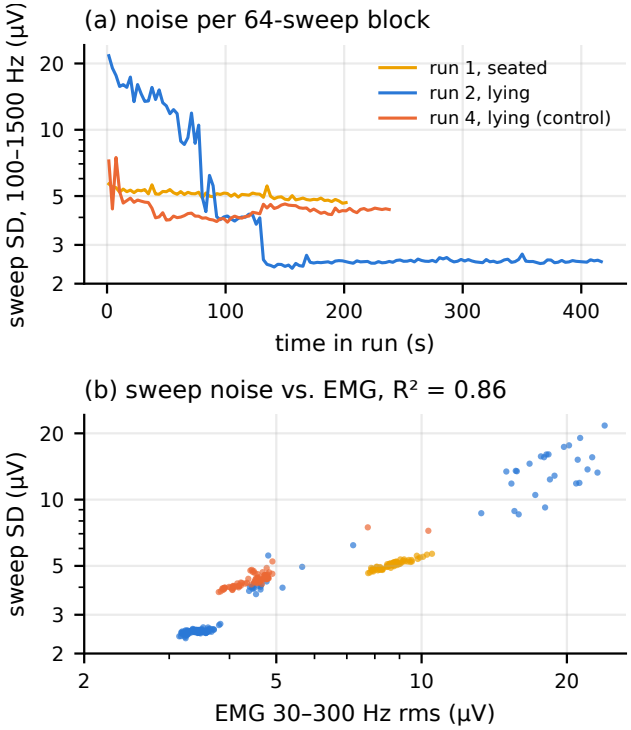


Figure 6: (a) Sweep noise over time. (b) Sweep noise against EMG (30 to 300 Hz) per 64-sweep block.

6.4 Other Signals in the Data

The same recordings contain more than the ABR. The forehead electrode picks up eye blinks (about 100 µV; 7 per minute seated, 1–2 per minute lying relaxed), and lying down with eyes closed shows a clear alpha rhythm at 10 Hz. The headset microphone shows every click as electrical crosstalk that scales exactly with the click amplitude, and the acoustic click only when the earbud is out of the ear. It can therefore serve as a stimulus monitor and as a check that the earbud is in place.

7 Limitations and Next Steps

Measurements. The ABR result rests on one subject, one evening and one significant run. The control run was noisier than run 2 and supports the result rather than proving it. The click level is not calibrated in dB. Next: a level series with interleaved earbud-out controls, a latency–intensity curve and a calibrated click.

Firmware. The 16-bit stream uses only 1.7 % of its range at gain 16; 24-bit samples or digital gain before truncation would remove the quantisation contribution. Exact click-onset markers in the packet header, a larger transmit buffer against the periodic Wi-Fi gaps and a live EMG gate that discards restless blocks would make measurements faster and more robust.

Hardware. For the next revision: a series resistor and clamp in front of the codec inputs (a saturated amplifier on ±5 V can exceed the codec's absolute maximum), I^2C pull-ups to the codec supply, the remaining open values of the electrode network, and the charger status routed to the MCU. The earphone cable should be twisted and led away from the electrode leads.

8 Conclusion

Wireless BERA v1.0 was designed, manufactured, assembled and brought into operation. On battery and over Wi-Fi it records a clean ECG that agrees beat for beat with a commercial device, its driven right leg lowers mains interference by up to 18 dB, and it recorded a click-evoked brainstem response with a wave-V candidate at 6.1 ms. The measurements also show where the remaining limit lies: in the muscle activity of the subject, not in the electronics.

Use of AI Tools

This report, the measurement scripts and parts of the firmware were drafted with the AI assistant Claude (Anthropic), working from the project files and the author's instructions; the author reviewed the results. The hardware design, assembly and measurements are the author's own work.

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